

TOKMAK, Kyrgyzstan

WMO: 369110

Lat: 42.833N

Lon: 75.283E

Elev: 2680

StdP: 13.33

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	7.0	-4.4	4.8	3.5	0.4	6.2	9.2	23.1	46.7	18.2	43.6	1.3	90	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	94.7	67.9	91.9	66.8	89.4	65.8	70.2	89.5	68.6	87.6	67.2	85.9	3.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	98.3	78.9	62.1	92.2	76.9	60.5	86.9	76.2	36.0	89.5	34.5	87.9	33.3	86.1	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	11.5	9.1	-5.2	99.5	7.1	2.0	-10.3	100.9	-14.4	102.1	-18.4	103.3	-23.5	104.8		
			-5.8	73.8	6.9	2.1	-10.7	75.3	-14.7	76.5	-18.5	77.7	-23.5	79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	28.9	32.2	44.9	55.3	64.0	72.2	76.1	74.3	64.6	53.4	41.6	31.4	
	DBStd	18.62	10.41	10.26	10.16	8.88	6.62	5.90	5.24	5.80	7.27	8.69	9.91	10.41	
	HDD50	2359	656	502	220	48	2	0	0	0	1	65	286	579	
	HDD65	5296	1120	919	623	307	98	12	1	6	95	368	705	1042	
	CDD50	3579	1	3	63	206	435	665	809	752	440	171	33	1	
	CDD65	1043	0	0	1	15	67	227	346	293	84	9	1	0	
	CDH74	11609	0	0	20	251	792	2422	3834	3161	976	142	11	0	
	CDH80	4840	0	0	1	67	220	979	1785	1425	331	31	1	0	
Wind	WSAvg	2.8	3.0	2.8	3.3	3.1	3.0	3.0	2.9	2.6	2.4	2.5	2.4	3.2	
Precipitation	PrecAvg	17.8	1.0	1.2	2.0	2.8	2.9	1.5	0.7	0.4	0.7	1.7	1.7	1.1	
	PrecMax	29.6	2.8	3.2	4.8	5.4	6.1	3.6	2.9	1.8	2.0	3.7	3.8	4.0	
	PrecMin	11.7	0.2	0.2	0.7	0.4	1.0	0.2	0.0	0.0	0.0	0.6	0.4	0.0	
	PrecStd	3.6	0.5	0.7	0.9	1.3	1.3	0.8	0.7	0.4	0.5	0.8	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	63.5	76.8	86.8	90.0	96.6	99.1	97.0	91.8	83.8	74.4	59.6	
		MCWB	42.3	45.5	53.1	60.7	63.9	66.9	69.5	69.3	64.9	58.5	53.5	44.3	
	2%	DB	51.4	56.3	71.7	80.9	85.5	92.2	95.5	93.8	87.5	77.7	66.4	54.0	
		MCWB	39.6	43.0	50.7	58.3	62.8	66.3	68.5	67.8	63.5	56.7	50.1	41.2	
	5%	DB	47.1	51.0	66.7	75.9	82.0	89.0	92.1	90.9	83.8	72.9	60.9	49.8	
		MCWB	37.0	40.5	49.3	55.8	61.7	65.4	67.4	66.6	62.0	54.4	47.5	39.3	
	10%	DB	43.1	46.6	61.4	71.3	78.4	85.8	89.3	88.0	79.8	68.2	55.4	45.9	
		MCWB	35.3	38.0	47.6	54.5	60.4	64.6	66.5	65.7	60.0	52.6	45.0	37.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	47.7	55.5	64.0	67.2	70.9	72.9	72.4	67.4	60.6	55.6	45.6
			MCDB	55.2	59.8	70.9	82.2	82.8	88.3	92.2	91.2	87.6	79.5	70.6	56.2
2%		WB	40.3	43.8	52.7	60.5	64.9	68.7	70.7	69.7	64.5	57.6	51.4	42.4	
		MCDB	49.5	54.3	67.4	75.0	80.4	86.1	90.3	89.8	84.3	74.3	64.2	51.8	
5%		WB	37.9	41.3	50.5	58.0	63.3	67.1	68.9	68.0	62.6	55.7	48.4	40.1	
		MCDB	45.9	49.6	63.5	71.4	78.5	84.3	88.4	87.6	81.6	69.6	58.7	48.4	
10%		WB	35.5	38.6	48.4	55.9	61.7	65.7	67.4	66.5	60.7	53.6	45.8	37.8	
		MCDB	42.2	45.9	59.8	68.1	76.2	82.6	86.6	85.5	78.1	65.9	54.1	45.3	
Mean Daily Temperature Range	5% DB	MDBR	18.2	18.2	19.8	21.0	22.5	22.9	23.0	24.2	24.9	23.4	19.2	17.9	
		MCDBR	22.3	24.4	27.8	29.0	27.5	26.6	26.2	26.5	28.8	30.2	26.9	22.4	
		MCWBR	13.2	13.6	12.8	12.8	12.3	11.9	11.7	12.0	13.3	14.0	13.6	13.0	
	5% WB	MCDBR	20.3	21.7	24.6	25.0	24.6	24.2	24.6	24.7	27.0	26.5	24.5	20.5	
		MCWBR	12.5	12.6	12.5	12.3	12.3	11.7	12.0	12.2	13.3	13.3	13.2	12.5	
Clear-Sky Solar Irradiance	taub	0.318	0.362	0.427	0.459	0.436	0.440	0.433	0.403	0.386	0.398	0.363	0.313		
	taud	2.179	2.085	1.987	1.970	2.120	2.156	2.195	2.268	2.255	2.107	2.131	2.197		
	Ebn at Noon	252	258	254	256	267	265	266	271	266	241	230	241		
	Edh at Noon	30	40	50	55	49	47	45	40	38	39	32	27		
All-Sky Solar Radiation	RadAvg	683	974	1270	1569	1936	2084	2105	1980	1594	1028	660	556		
	RadStd	49	73	95	130	146	104	126	106	79	106	62	38		

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(a)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(b)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.86	-1.58	N/A	N/A	N/A

Nomenclature: See separate page